Cristian Cisneros

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Research interests: nonlinear dynamics, fluid dynamics, multifractal analysis, machine learning for dynamical systems

Education

University of Houston Houston, TX
B.S., Physics and Mathematics (double major) *Expected May 2027*
GPA: 3.46/4.00

- **Graduate coursework:** Differential Equations (MATH 6324); Partial Differential Equations (MATH 6326, in progress); Advanced Mechanics I (PHYS 6319, in progress); Electrodynamics (PHYS 6321, Spring 2027)
- **Selected upper-level coursework:** Classical Mechanics; Electromagnetic Theory I/II; Quantum Mechanics; Thermal Physics; Theory of ODE and Nonlinear Dynamics

Research Experience

Undergraduate Researcher, Nonlinear Dynamics Nov 2025 – Present
Department of Physics, University of Houston *Advisor: Dr. G. Gunaratne*

- Processed the human genome (UCSC hg38) into left- and right-weighted quaternary expansions of nucleotide sequences and compared exons and introns through multifractal analysis (published; see [1]).
- Generated gene-specific point-set representations of exons and introns and trained a CNN that classifies them as exon or intron with 97% accuracy.
- Performed multifractal analysis of strange attractors, primarily the Hénon map, and trained CNNs to distinguish different parameter regimes and associated invariant measures, using torus automorphisms for data augmentation.
- Probe which dynamical invariants the network learns by comparing its outputs against generalized dimensions, Lyapunov exponents, and correlation measures.
- Wrote multifractal analysis pipelines (box-counting $\tau(q)$ and $D(q)$, Legendre transform to $f(\alpha)$, Badii–Politi nearest-neighbor estimator) and computed Lyapunov spectra by QR iteration with Kaplan–Yorke consistency checks.
- Ran training and parameter sweeps on UH's Carya and Sabine HPC clusters via Slurm.

Research Intern, Frontiers in Science Program Summer 2025
Center for Theoretical Biological Physics, Rice University *PI: Dr. José N. Onuchic*

- Simulated folding of an mRNA riboswitch across ion concentrations using structure-based molecular dynamics models.
- Ran simulation campaigns on Rice's NOTS cluster with Slurm and MPI.
- Presented results as a poster in August 2025.

Publications

[1] J. Shulman, **C. M. Cisneros**, P. H. Gunaratne, and G. H. Gunaratne, "Classifying Consensus Sequences Using Point-Set Representations," *Mathematics* **14**, 1826 (2026). doi:10.3390/math14111826

Presentations

- **C. Cisneros**, E. Ding, and J. N. Onuchic, "Computational Analysis of mRNA Riboswitch Folding at Varying Ion Concentrations," poster, FIS Poster Day, Rice University, Houston, TX, Aug. 2025.

Awards and Funding

Undergraduate Research Support, Department of Physics, University of Houston 2026

Teaching and Service

Peer Facilitator Fall 2026 – Present
University of Houston *College Physics I*

- Lead workshop sessions demonstrating problem solving in physics and help students with fundamental topics.

Manager, Math Modeling Club (Cougar Mathletics) Spring 2026 – Present
University of Houston *Faculty advisor: Dr. G. Jaramillo*

- Run outreach for weekly math-modeling talks and recruit for the SCUDEM modeling challenge; competing in SCUDEM XI (Fall 2026).

Martial Arts Instructor Apr 2020 – Present
USA Karate *Houston, TX*

Skills

Programming: Python (NumPy, TensorFlow), MATLAB, LaTeX
High-performance computing: Slurm, MPI; UH Carya and Sabine, Rice NOTS clusters
Visualization: Tableau

Certifications

HPE Data Science Institute (UH): Cluster Computing, Scientific Programming with Python, Data Analysis in R, Data Visualization, Intro to Machine Learning 2026